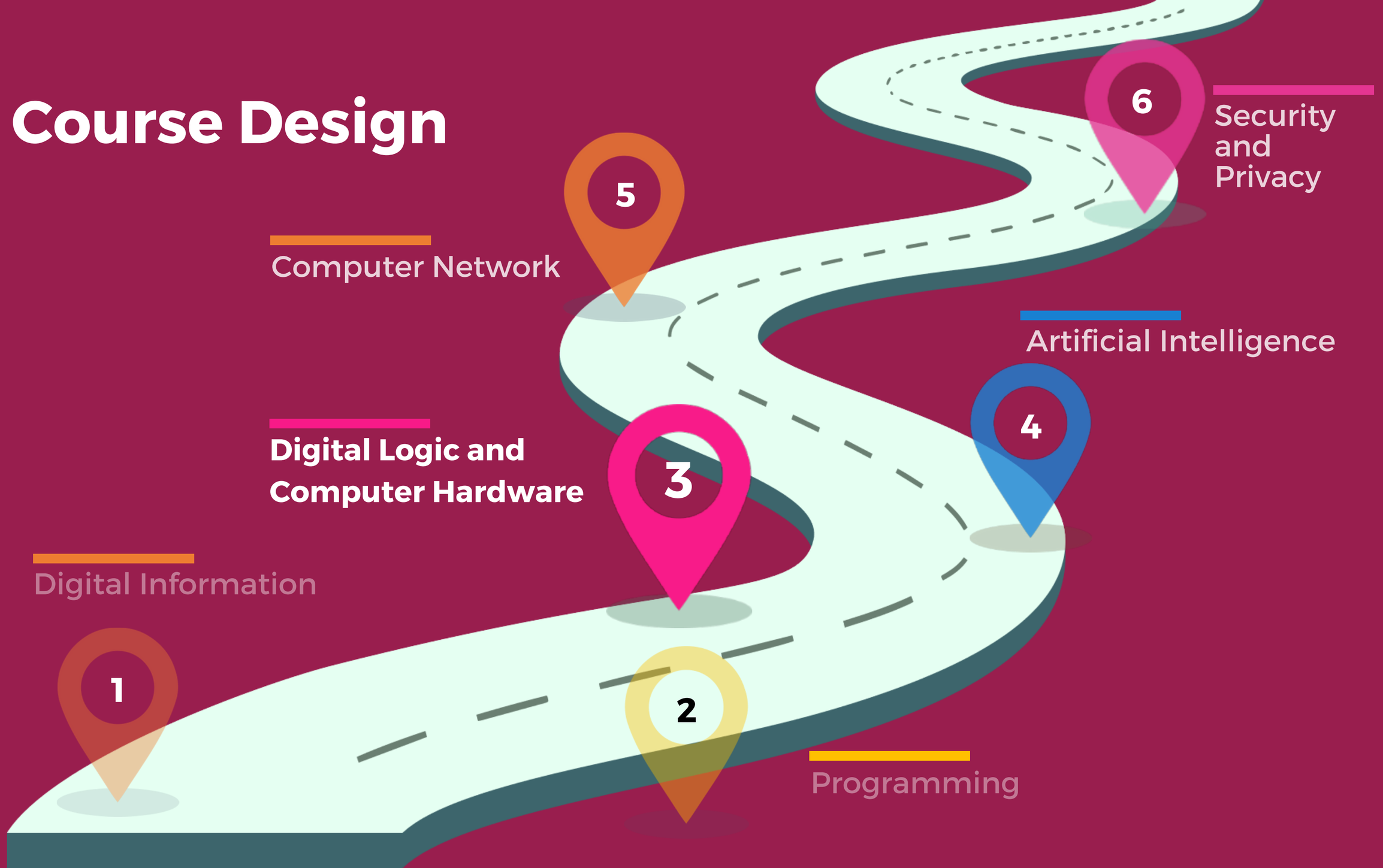
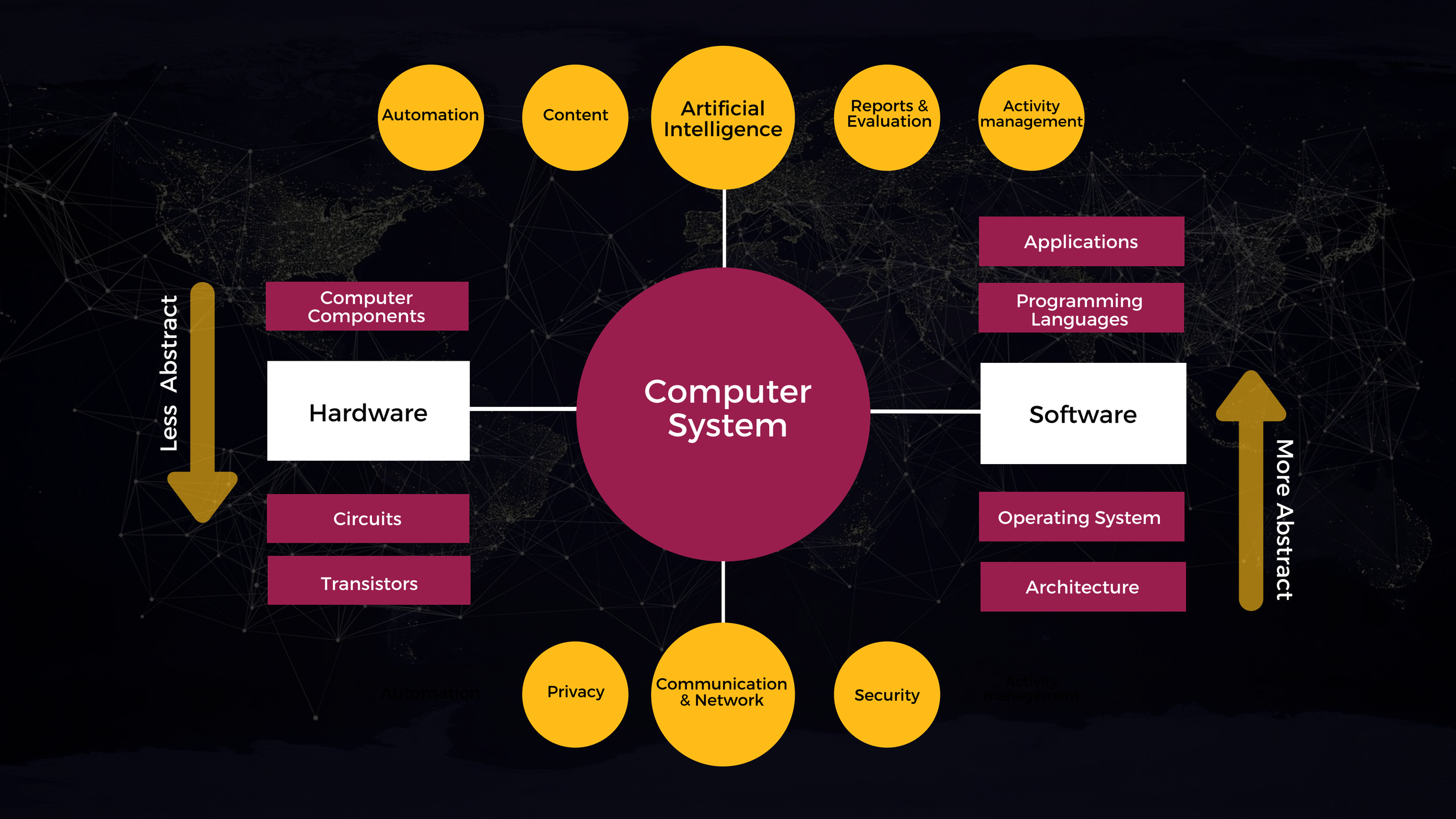
LOYOLA UNIVERSITY CHICAGO

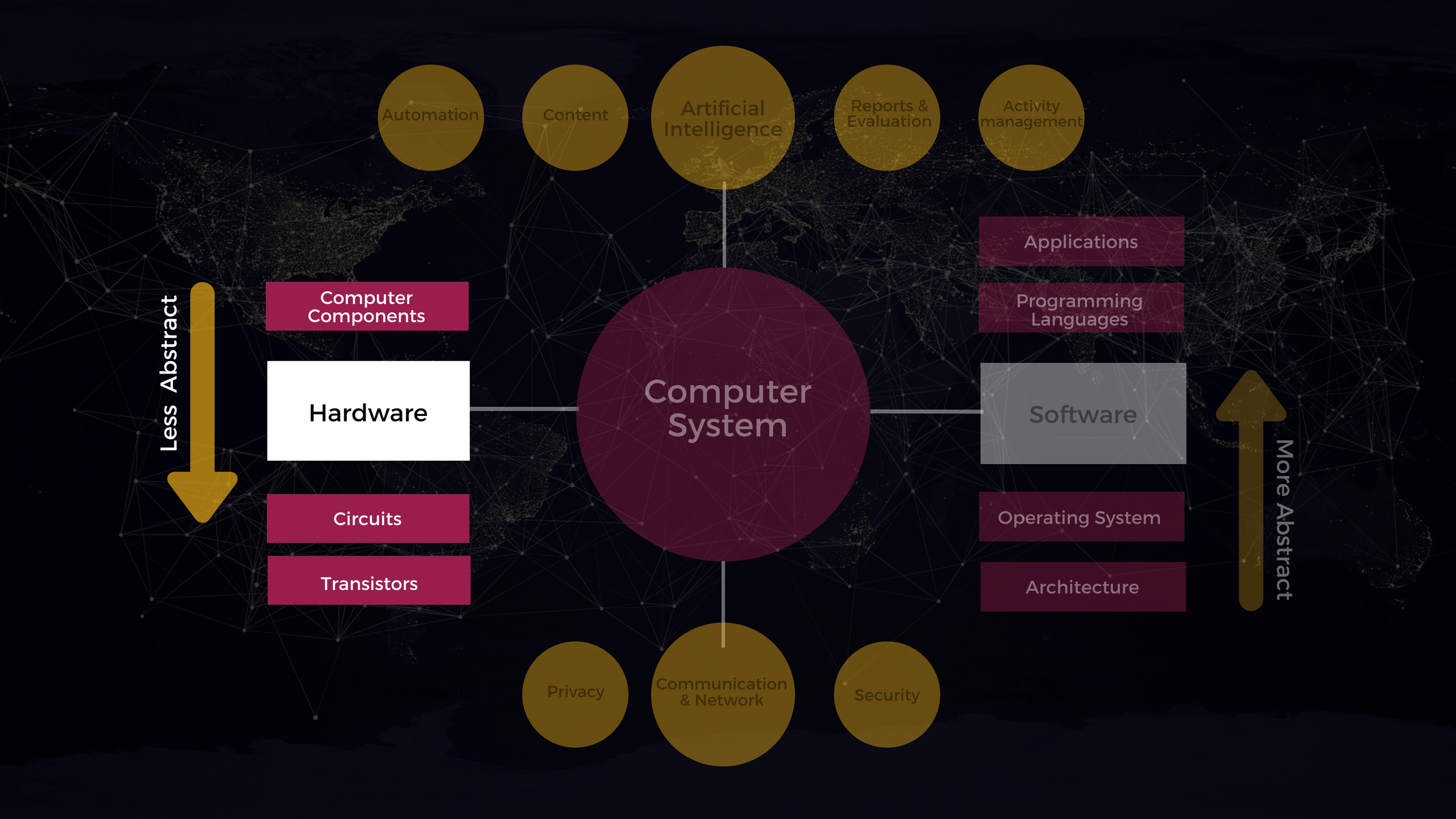
Introduction to Computing

MODULE 3: DIGITAL LOGIC AND COMPUTER HARDWARE

Course Design




Automation
Content
Artificial Intelligence
Reports & Evaluation
Activity management
Computer Components
Hardware
Circuits
Transistors
Less Abstract
Computer System
Applications
Programming Languages
Software
Operating System
Architecture
More Abstract
Privacy
Communication & Network
Security


Automation
Content
Artificial Intelligence
Reports & Evaluation
Activity management
Computer Components
Hardware
Circuits
Transistors
Less Abstract
Computer System
Applications
Programming Languages
Software
Operating System
Architecture
More Abstract
Privacy
Communication & Network
Security

Computer Hardware

- Hardware is the most visible part of any computer system.
- Computer hardware is all equipment that is used for:
  - Capturing data
  - Processing it
  - Storing it
  - Presenting it to the user.

Computer Hardware

- Hardware equipment includes:
  - Input devices:
    - Keyboard, mouse, camera, touchpad, scanners, etc.
  - Processing units
  - Storage units and memory
  - Output devices:
    - Screens, printers, etc.

Computer Hardware

- This module focuses on explaining the hardware components of personal computers (PC). However, the same concepts extend to the entire range of computers, such as:
  - Supercomputers
  - Mainframe computers
  - Servers
  - Workstations and personal computer
  - Mobile devices
  - IoT devices

SUPERCOMPUTERS

- Provide fast and large computing engines for running highly-demanding scientific applications.

MAINFRAME COMPUTERS

- Provide high processing and data storage capabilities for a large number of users.

SERVERS

- Provide large data storage capacities and allow users to share data and software applications. Typically, the processing happens locally on the client machine.

WORKSTATIONS AND PERSONAL COMPUTERS

- This type (including personal machines and laptops) is the one you know so well. PCs provide convenient processing units that can operate independently or as part of large networks.

MOBILE DEVICES

- Provide cellular services and wireless connection to the internet. Currently, mobile devices are capable of providing a wide range of functionalities.

IOT DEVICES

- Provide resources (computation units, sensors, etc.) to conduct specific tasks. IoT devices include wearable devices, embedded devices in cars, house appliances, healthcare devices, traffic controllers, etc.

Computer Hardware

- Hardware equipment includes:
  - **Input devices:**
    - **Keyboard, mouse, camera, touchpad, scanners, etc.**
  - Processing units
  - Storage units and memory
  - Output devices:
    - Screens, printers, etc.

INPUT DEVICES

Input Devices

- Data may enter an information system in a variety of different ways.
- Choosing the appropriate input device depends on the type of data.
  - Typing/scanning
  - Video/audio recording
  - Image/text postings
  - Others
- All input devices have something in common.
  - The ability to translate non-digital data into their binary format.

Input Devices: The Keyboard

- Typing textual content happens using a keyboard.
  - Actual physical keyboard that has a similar layout as a typewriter.
  - On-Screen Keyboard that allows for touch input on mobile devices.
- The most common keyboard layout is QWERTY.
  - QWERTY was originally designed for typewriters and meant to slow down the writer so the keys would not get stuck against each other.
  - We still use this layout, even though the current keyboards can process information multiple times faster than the fastest typist.

Input Devices: The Keyboard

- There are many designs and layouts available nowadays to allow for:
  - Faster input
  - Other languages
  - Input from users with disabilities.

Input Devices: Pointing Devices

- **Mouse**
  - Movements are translated to electronic signals that allow the cursor to correspond to the movement on the computer monitor
- **Trackball**
  - It acts like an upside-down mouse with a larger rolling-action ball. Instead of moving the mouse, the user can direct the cursor by moving the ball directly.

Input Devices: Pointing Devices

- **Touch-screen/touch-pad**
  - Touch-screens are monitors with sensors on the screen panel.
  - Users can move or indicate a selection by pressing on the screen.
  - Special pens can be used to trace the movement or to support typing (with handwriting recognition) and drawing.
  - Touch-pads are different and come with most laptops.
    - They are meant to direct the cursor by the movement of the fingers on the pad.

Input Devices: Pointing Devices

- Many other devices are used for games and virtual reality.
  - Such as joysticks, steering wheels, data glove, etc.
  - All use the same concepts to translate the movements to electronic signals that provide real-time motion control.

Input Devices: Optical scanners and readers

- There are several technologies for optical scanners on the market.
  - Normal optical scanners
  - Barcode scanners
  - Optical mark reader

Input Devices: Optical scanners and readers

- **Barcode scanners:**
  - Detect sequences of vertical lines with various widths.
    - Very popular with retailers.
      - All packaged products are required to have a product bar code.
  - They are used for tracking large numbers of physical items.
    - Postal items, luggage, books, etc.

Input Devices: Optical scanners and readers

- **Optical mark readers:**
  - They are designed to read dark marks on special forms.
  - They help with reading the marks of answer sheets in many educational and testing centers.

Input Devices: Other Input Devices

- **Magnetic card readers:**
  - They are capable fo reading the magnetized stripe on plastic cards
    - Credit cards, university ID cards, etc.
  - Can be found in access control points (e.g., doors), ATMs, point-of-sale counters

Input Devices: Other Input Devices

- **Magnetic ink character recognition (MICR):**
  - Uses special ink (with magnetizable elements) and fonts.
  - Used by the banking industry for printing and reading cheques.

Input Devices: Other Input Devices

- **Digital cameras:**
  - Capture visual information and translate it into digital format.
  - Most of today's devices (personal computers and mobile phones) are equipped with high-resolution cameras.

Input Devices: Other Input Devices

- **Biometric devices:**
  - Such as fingerprint readers, Iris scanners, hand geometry scanners.
  - Such devices are used for identity verification.
  - There is a trend of using biometric authentication in most personal devices.
  - Biometric-based systems/readers are capable of capturing predefined characteristics from the target biometric data and compare it to pre-stored templates.

Input Devices: Other Input Devices

- **Voice input devices (microphones):**
  - Microphones are used for audio recording and voice interactions.
  - Voice control and voice recognition allow users to communicate with others as well as smart systems (e.g., personal assistants (Google Assistant, Siri, Bixby, Alexa, etc.) and voice recognition).

Computer Hardware

- Hardware equipment includes:
  - Input devices:
    - Keyboard, mouse, camera, touchpad, scanners, etc.
  - **Processing units**
  - Storage units and memory
  - Output devices:
    - Screens, printers, etc.

CENTRAL PROCESSING UNIT (CPU)

Central Processing Unit (CPU)

- Once the data is inputted to the system, the CPU applies a set of instructions to the data to generate an output.
- The CPU is based on microchip technology and it is the showcase of the electronics industry.
  - Hence it is also called the microprocessor.
- One square cm contains several millions of transistor elements

Central Processing Unit (CPU)

- Companies like Intel are mass-producing transistors 14 nanometers across.
  - 14 times wider than DNA molecules
- They’re made of silicon.
  - The second-most abundant material on our planet.
  - Silicon’s atomic size is about 0.2 nanometers.
  - Today’s transistors are about 70 silicon atoms wide
- They are very fast.
  - A regular CPU can multiply more 7-digit numbers in one second than a human could do in ten lifetimes.

Central Processing Unit (CPU): Components

Components of the CPU:

- The CPU has two major components.
  - **1. The Arithmetic and Logic Unit (ALU)**
    - Executes instructions.
      - Add/multiply numbers, compare data, or transform data into different formats.
  - **2. The Control Unit**
    - Ensures the proper processing of instructions.
      - Processing instruction on time, in the right sequence, and on the correct data.

Central Processing Unit (CPU): Types

Types of CPUs

- The best-known manufacturers are Intel and AMD.
  - Each produced three series of processors.
    - Extreme or high-end line
    - Mainstream line
    - Budget line.
  - There are many different types of CPUs on the market.
    - Different speed, architecture, number of cores, etc.

Central Processing Unit (CPU): Speed

Speed of processing

- The speed of the CPU can be measured by the time it takes to process a single instruction.
  - An instruction can be done in about two billionths of a second (i.e., 2 nanoseconds).
- However, the processing speed of a CPU is usually calculated by the number of instructions a CPU can execute in a second.

Central Processing Unit (CPU): Speed

Speed of processing

- Let's say a CPU takes 2 nanoseconds per instruction.
  - Then, it can execute 500 million instructions per second.
    - 1/(2/1000,000,000) --- ?
- The rate at which the CPU can process instructions is called the **clock rate**.
  - For the CPU with 500 million instructions per second
    - The clock rate is 500 megahertz (MHz)
- Today's CPUs have a clock rate in GHz.

Central Processing Unit (CPU): Speed

Speed of processing

- Mainframes and supercomputers use more complex measurements for the speed of the processing unit.
    - For example, **megaflops** (a million of floating-point operations per second) or gigaflops (a billion of floating-point operations per second)

Central Processing Unit (CPU): Speed

Speed of processing

- The speed of a processor is dictated by four different elements:
  - **Clock speed:**
    - Instructions/seconds
  - **The word length:**
    - The number of bits to be processed at any one time.
  - **The bus width:**
    - The number of bits that can be transferred concurrently in or out of the CPU.
  - **The physical design of the chip.**

Computer Hardware

- Hardware equipment includes:
  - Input devices:
    - Keyboard, mouse, camera, touchpad, scanners, etc.
  - Processing units
  - **Storage units and memory**
  - Output devices:
    - Screens, printers, etc.

STORAGE UNITS AND MEMORY

Storage Units and Memory

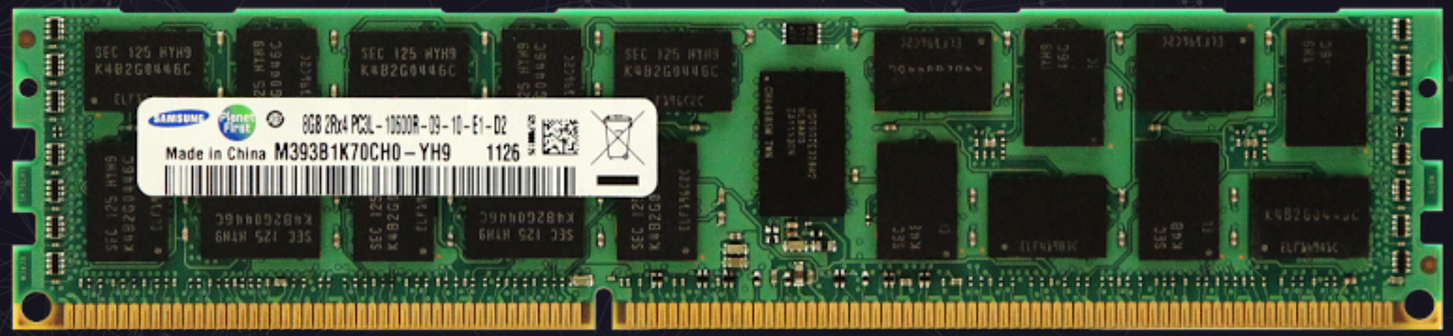
- Main Memory - Random Access Memory (RAM)
- Read-Only Memory (ROM)
- Secondary Storage Devices

Storage Units and Memory: Main Memory

- The main memory is also referred to as primary memory, main storage, or internal storage.
  - Known as Random Access Memory (RAM)
- It provides temporary storage for instructions and data during the execution time.
- RAM consists of standard circuit-inscribed silicon microchips that contain many millions of tiny transistors.

Storage Units and Memory: Main Memory

- RAM are frequently pre-soldered in sets on tiny memory circuit boards called SIMMS (Single In-line Memory Modules) or DIMMS (Dual In-line Memory Modules).



- The biggest disadvantage of RAM is that its contents are lost whenever the power is switched off.

Storage Units and Memory: Main Memory

- There are many special types of RAM:
  - **Cache memory:**
    - Ultra-fast memory that operates at the speed of the CPU.
    - Sits in between the CPU and RAM to allow fast processing.
    - It holds the data and programs that are needed immediately by the CPU.
  - **Flash RAM or flash memory:**
    - Unlike normal RAM, flash memory is non-volatile, i.e. it holds its contents even without external power, so it is also useful as a secondary storage device.

Storage Units and Memory: Read-Only Memory

Read-Only Memory (ROM)

- A small but essential element of any computer.
- Unlike RAM, it does not lose its contents when the power is switched off
- Since it is difficult or impossible to change the contents of ROM, it is typically used to hold program instructions that are unlikely to change during the lifetime of the computer.
- The main application of ROM is to store the so-called **Boot Program**:
  - The instructions that the computer must follow just after it has been switched on.
    - Perform a self-diagnosis
    - Load the operating system from secondary storage.

Storage Units and Memory: Read-Only Memory

ROM comes in several different forms:

- **PROM** (Programmable Read-Only Memory):
  - It is initially empty and can be custom-programmed once only using special equipment.
- **EPROM** (Erasable Programmable Read-Only Memory):
  - It is like PROM but, by using special equipment such as an ultra-violet light gun, the memory contents can be erased so that the EPROM can be re-programmed.
- **EEPROM** (Electrically Erasable Programmable Read-Only Memory):
  - It is similar to EPROM but it can be re-programmed using special electronic pulses rather than ultraviolet light so no special equipment is required.

Storage Units and Memory: Secondary Storage

- Since the main memory of a computer has a limited capacity, it is necessary to retain data in secondary storage between different processing cycles.
- This is the medium used to store the program instructions as well as the data required for future processing.
- Most secondary storage devices in use today are based on magnetic or optical technologies

Storage Units and Memory: Secondary Storage

- Since the main memory of a computer has a limited capacity, it is necessary to retain data in secondary storage between different processing cycles.
- This is the medium used to store the program instructions as well as the data required for future processing.
- Most secondary storage devices in use today are based on solid state, magnetic, or optical technologies

Storage Units and Memory: Secondary Storage

- **Solid state storage:**
  - Special type of storage made from silicon microchips.
  - It can be written to and overwritten like RAM.
  - Unlike RAM, it is non-volatile and the solid state storage will retain its contents after the power is switched off.
  - Examples:
    - Solid State Drive (SSD)
    - Flash Drive
  - Advantages:
    - No Moving parts (i.e., less heat, portable, and durable)
    - Fast

Storage Units and Memory: Secondary Storage

- **Magnetic storage devices:**
  - Magnetic devices such as Hard Disk Drives (HDD) use magnetic fields to magnetise tiny individual sections of a metal spinning disk.
  - Each tiny section represents one bit.
    - A magnetised section represents a binary '1'.
    - A demagnetised section represents a binary '0'.
  - HDD can contain terabytes (TB) of data.
  - As the disk is spinning, a read/write head moves across its surface.
  - Advantages:
    - Cheap
    - High in capacity
    - Durable

Storage Units and Memory: Secondary Storage

- **Optical storage devices:**
  - Optical devices use a laser to scan the surface of a spinning disc.
  - The disc surface is divided into tracks.
    - Each track containing many flat areas and hollows.
      - The flat areas are known as lands and the hollows as pits.
  - When the laser shines on the disc surface:
    - Lands reflect the light back.
      - A sensor reads reflected light as a binary '1'.
    - Pits scatter the laser beam. A sensor looks for the reflected light.
      - A sensor reads reflected light as a binary '0'.

Computer Hardware

- Hardware equipment includes:
  - Input devices:
    - Keyboard, mouse, camera, touchpad, scanners, etc.
  - Processing units
  - Storage units and memory
  - **Output devices:**
    - **Screens, printers, etc.**

OUTPUT DEVICES

Output Devices

- The final stage of information processing involves the use of output devices to transform computer-readable data back into an information format that can be processed by humans.
  - Display screens
  - Printers and plotters
  - Audio-output devices
  - Other Output Devices

Output Devices: Display Screens

- **Cathode-Ray Tube (CRT):**
  - A “ray” gun fires electrically charged particles onto a specially coated tube.
  - The particles hit the coating, the “coating” is being “excited” and emits light.
  - A strong magnetic field guides the particle stream to form the text or graphics on your familiar monitor.

Output Devices: Display Screens

- **Liquid crystal display (LCD):**
  - LCDs have matured quickly, increasing in resolution, contrast, and color quality.
  - Their main advantages are lower energy requirements and their thin, flat size
- **Organic light-emitting diodes (OLED):**
  - Generate brighter and faster images than LED technology
  - Require thinner screens, but they have less stable color characteristics.
  - More suitable for phones than for computers.

Output Devices: Display Screens

- **Projection units:**
  - Used for computer-driven public presentations

Output Devices: Printers and Plotters

- Types of printers.
  - Dot-matrix printers
    - Low-cost printer
    - Print head consists of a vertical row of needles each controlled by a magnet.
    - The print head moves horizontally across the paper and different needles hit the paper through a ribbon (between the paper and the print head) to create text characters or graphics.

Output Devices: Printers and Plotters

- Types of printers.
  - Laser printers
    - A laser beam lights on and off very quickly, illuminating selected areas on a photo-sensitive drum
    - The light is then transformed into an electrical charge.
    - The drum rotates into a “bed” of toner (i.e., carbon particles with the opposite charge, so they adhere to the drum.
    - The paper is pressed against the drum and the toner is attached to the paper.
    - The paper then goes through a high-temperature area so that the toner is permanently printed on the paper.

Output Devices: Printers and Plotters

- Types of printers.
  - Thermal printers
    - The thermal printers use heat-sensitive paper.
      - Similar to the special fax paper.
    - Heat or pressure leaves darker areas (e.g., old thermal preinters).
    - Nowadays, thermal-printing technology use colored wax sticks to melt colors on paper for high-quality color printing.

Output Devices: Printers and Plotters

- Types of printers.
  - Inkjet printers
    - Popular and low-cost printing technology.
    - Liquid ink is emitted onto the paper.
      - As tiny droplets.
    - Similar quality as laser printers.
    - Inkjet printers use colored ink for color printing.

Output Devices: Printers and Plotters

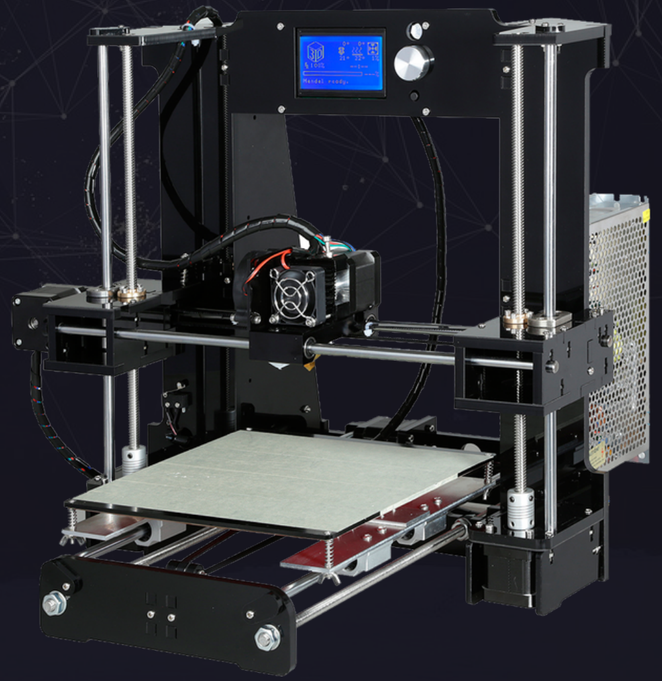
- Types of printers.
  - Plotters
    - Plotters are used for engineering and architectural drawings.
    - A plotter consists of one pen (for one-colored plots) or multiple pens (for colored plots) attached to an arm.
    - The arm moves and draws lines across the paper.
    - Plotters are ideal for line drawings on larger paper size than the one accommodated by typical printers.

Output Devices: Printers and Plotters

- Types of printers.
  - Chain and line printers
    - This type of printers is used to be popular in large organizations for quick and large internal printing.
    - The printer has a horizontal rotating “drum” with 132 cylinders.
      - Each cylinder has a full character set.
    - As the paper moves against the drum, one line at a time, each one of the 132 cylinders strikes simultaneously with the correct character.
    - After the cylinder-hammers strike the paper and the ink ribbon, they leave the printed character on the paper.

Output Devices: Printers and Plotters

- Types of printers.
  - 3D printers
    - They are used to construct a three-dimensional object from a digital 3D model.

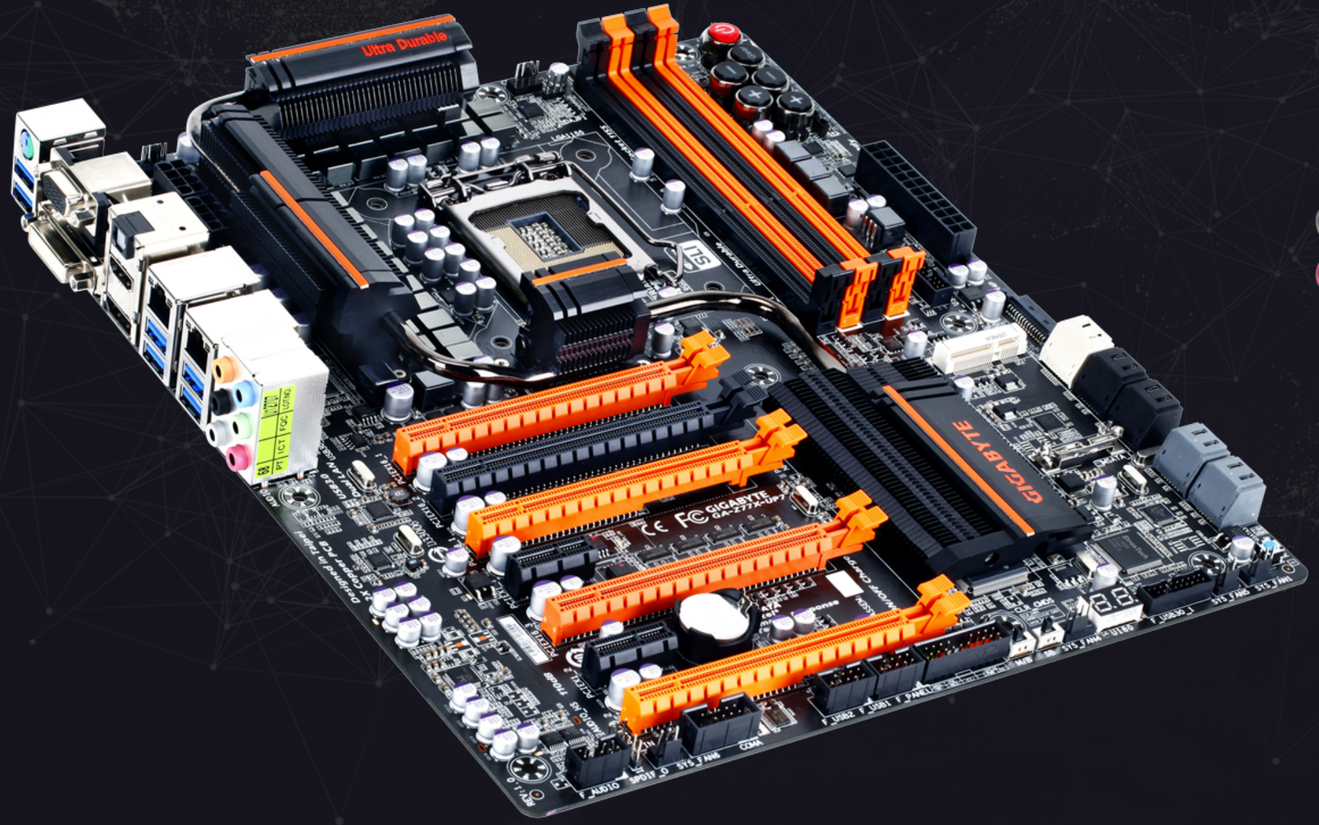


A LOOK INSIDE THE COMPUTER

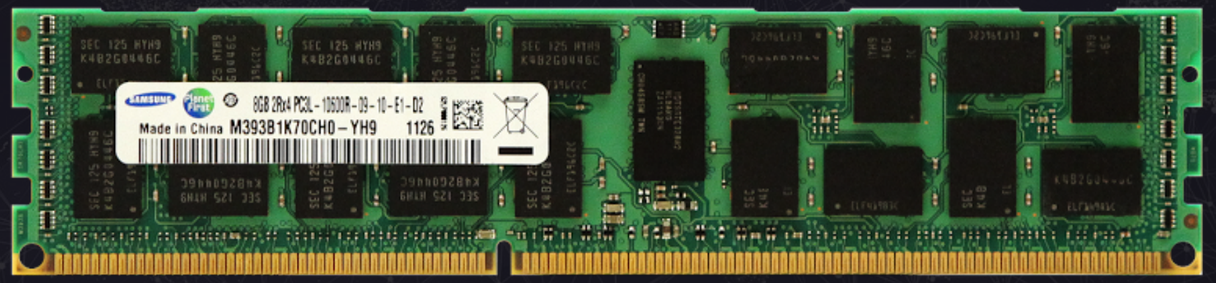
HYPERX
FURY
DDR4
AX1500i

1
2
3
4
5
6
7
8
HYPERX
FURY
DDR4
AX1500i

1 Motherboard



2 RAM

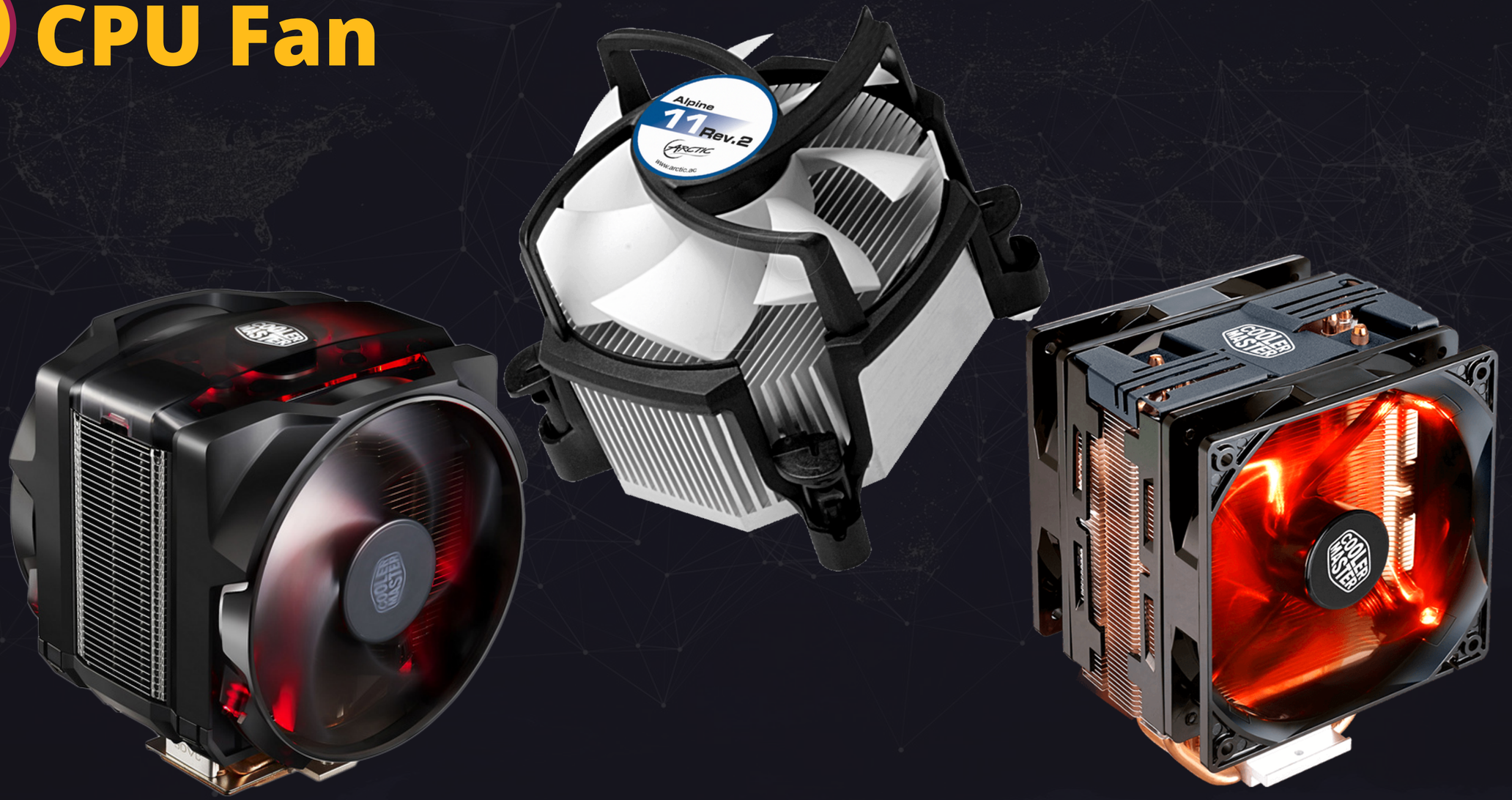

SAMSUNG
8GB 2Rx4 PC3L-10600R-09-10-E1-D2
Made in China M393B1K70CH0-YH9 1126

HYPERX
PREDATOR
DDR4

3 CPU

intel
Core™2 Duo
intel
Core™ i7
AMD
RYZEN
intel
Xeon® E3-1200
intel
CORE i9
X-series
intel
Xeon®

4 CPU Fan


Alpine
11 Rev.2
ARCTIC
www.arctic.ac
COOLER MASTER
COOLER MASTER
COOLER MASTER

5 HDD

Internal Hard Disk
Handle the hard drive carefully to avoid damaging the circuit board.
Make sure you are properly grounded.

Western Digital
www.westerndigital.com
3 TB
SATA / 64MB Cache
WD30EZRS
WD Caviar Green

6 SSD

SAMSUNG
Solid State Drive
NVMe SSD
960 PRO
SAMSUNG
2TB
crucial
by micron
2.5-inch SOLID STATE DRIVE
SSD

7 GPU

8 Power Supply

End Part 1